



FutureXcel Internship Program

Title:

Feature Engineering & Transformation Using Superstore Dataset

Role:

Data Science Intern

Organization:

FutureXcel

Week: 4

Feature Engineering & Transformation Suite

Prepared by:

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Tools & Technologies:

Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Jupyter Notebook

Submission Date:

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1. Introduction

Feature Engineering is an important step in the data science workflow. It helps improve the performance of machine learning models by creating meaningful variables from raw data.

In this task, the **Sample - Superstore** dataset was used to create new features, transform existing variables, and prepare the dataset for analysis and modeling.

The objective of this task is to:

- Create new meaningful features
- Apply encoding techniques
- Apply feature scaling
- Analyze how transformations improve the dataset

2. Dataset Overview

The Superstore dataset contains retail transaction data from a fictional store.

Main information included in the dataset:

- Orders
- Products
- Customers
- Sales
- Profit
- Shipping details

Important columns include:

- Order Date
- Ship Date
- Sales
- Profit
- Quantity
- Discount
- Category
- Region
- Segment

These columns allow us to perform feature engineering and data transformation.

Task4.ipynb > M4 Step 10 — Save Final Dataset > empty cell

Python 3.10.11

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	Postal Code	Region	Prod
0	1	2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	42420	South FUR-I 10001
1	2	2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	42420	South FUR-I 10000
2	3	2016-138688	6/12/2016	6/16/2016	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	90036	West OFF-10000
3	4	2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	33311	South FUR-10000
4	5	2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	33311	South OFF-10000

5 rows x 21 columns

Task4.ipynb > M4 Step 10 — Save Final Dataset > empty cell

Python 3.10.11

Customer Segment	Country	City	Postal Code	Region	Product ID	Category	Sub-Category	Product Name	Sales	Quantity	Discount	Profit
Consumer	United States	Henderson	42420	South	FUR-BO-10001798	Furniture	Bookcases	Bush Somerset Collection Bookcase	261.9600	2	0.00	41.9136
Consumer	United States	Henderson	42420	South	FUR-CH-10000454	Furniture	Chairs	Hon Deluxe Fabric Upholstered Stacking Chairs...	731.9400	3	0.00	219.5820
Corporate	United States	Los Angeles	90036	West	OFF-LA-10000240	Office Supplies	Labels	Self-Adhesive Address Labels for Typewriters b...	14.6200	2	0.00	6.8714
Consumer	United States	Fort Lauderdale	33311	South	FUR-TA-10000577	Furniture	Tables	Bretford CR4500 Series Slim Rectangular Table	957.5775	5	0.45	-383.0310
Consumer	United States	Fort Lauderdale	33311	South	OFF-ST-10000760	Office Supplies	Storage	Eldon Fold 'N Roll Cart System	22.3680	2	0.20	2.5164

The screenshot shows a Jupyter Notebook titled 'Task4.ipynb' in a VS Code environment. The notebook is at 'Step 2 — Load Dataset' and displays the output of a `df.info()` command. The output shows a DataFrame with 9994 entries and 21 columns. The columns and their data types are as follows:

#	Column	Non-Null Count	Dtype
0	Row ID	9994 non-null	int64
1	Order ID	9994 non-null	object
2	Order Date	9994 non-null	object
3	Ship Date	9994 non-null	object
4	Ship Mode	9994 non-null	object
5	Customer ID	9994 non-null	object
6	Customer Name	9994 non-null	object
7	Segment	9994 non-null	object
8	Country	9994 non-null	object
9	City	9994 non-null	object
10	State	9994 non-null	object
11	Postal Code	9994 non-null	int64
12	Region	9994 non-null	object
13	Product ID	9994 non-null	object
14	Category	9994 non-null	object
15	Sub-Category	9994 non-null	object
16	Product Name	9994 non-null	object
17	Sales	9994 non-null	float64
18	Quantity	9994 non-null	int64
19	Discount	9994 non-null	float64
20	Profit	9994 non-null	float64

The output also indicates the dtypes: `float64(3), int64(3), object(15)` and the memory usage: 1.6+ MB. The Explorer sidebar on the left shows files: 'Sample - Superstore.csv', 'superstore_feature_engineered.csv', and 'Task4.ipynb'. The bottom status bar shows 'Spaces: 4', 'Cell 5 of 60', and the date '2/24/2026'.

3. Data Preparation

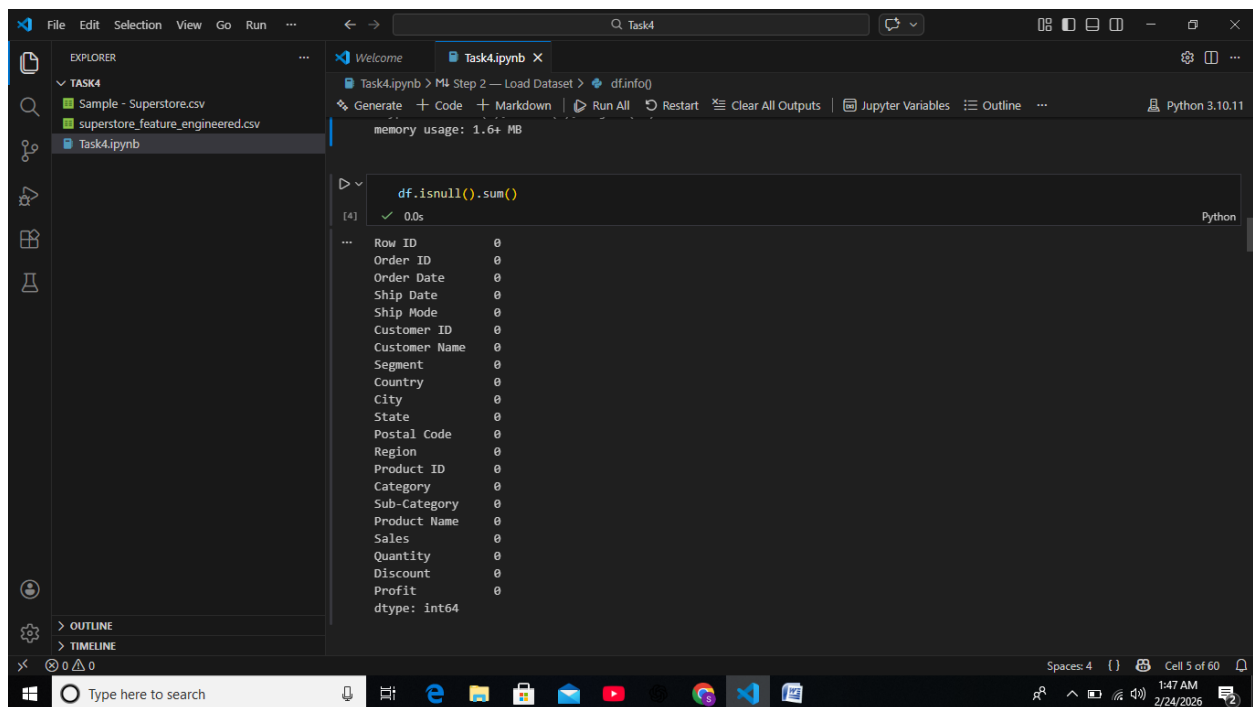
Before creating new features, the dataset must be cleaned and prepared.

The following steps were performed:

- Converted date columns to datetime format
- Checked for missing values
- Verified dataset structure

```
df['Order Date'] = pd.to_datetime(df['Order Date'])  
df['Ship Date'] = pd.to_datetime(df['Ship Date'])
```

```
df.isnull().sum()
```



4. Feature Engineering

Feature engineering was performed to create new meaningful variables from existing columns.

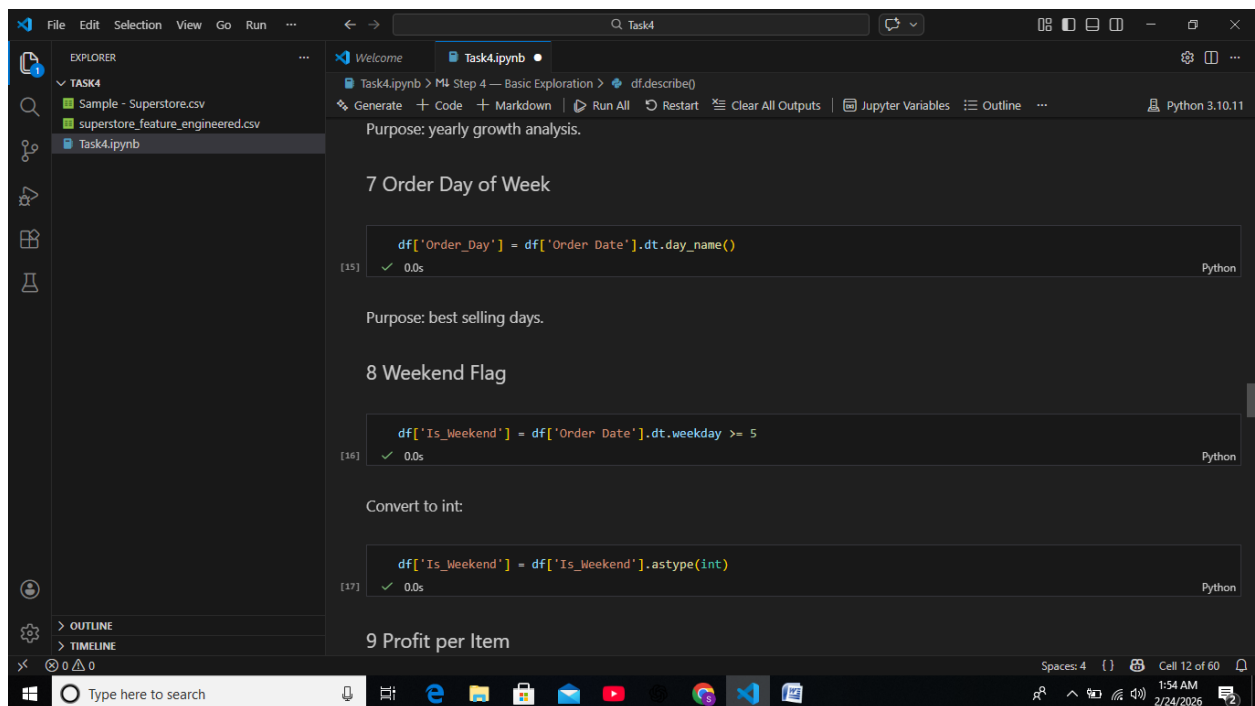
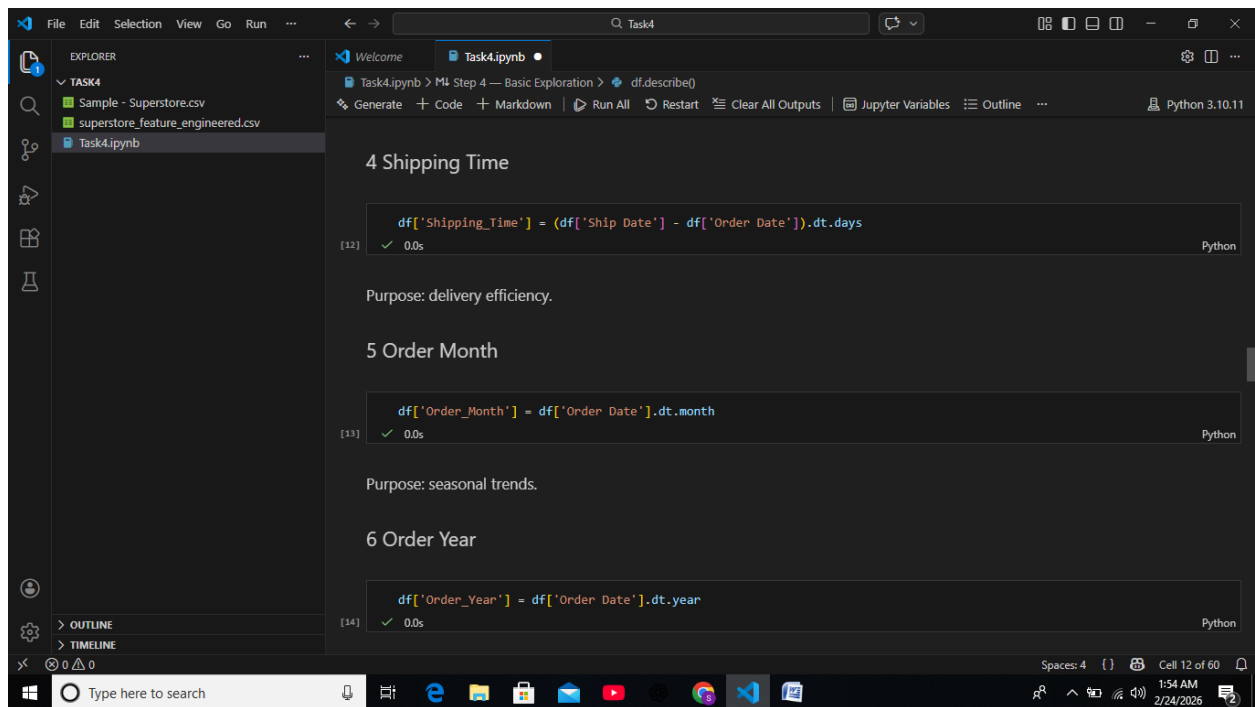
The following features were created:

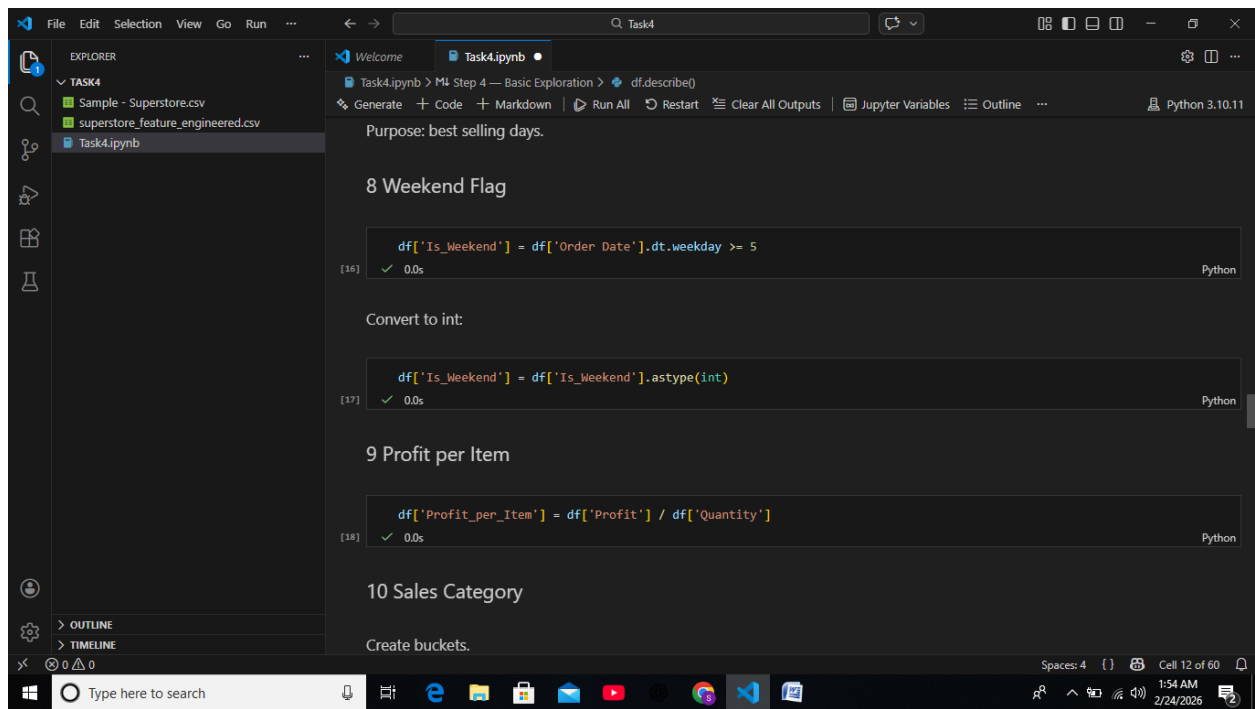
Feature	Formula	Purpose
Profit_Ratio	Profit / Sales	Measure profitability
Sales_per_Item	Sales / Quantity	Estimate item price
Discount_Amount	Sales $\times$ Discount	Calculate real discount
Shipping_Time	Ship Date – Order Date	Delivery performance
Order_Month	Month(Order Date)	Seasonal trends
Order_Year	Year(Order Date)	Annual analysis
Order_Day	Day name	Identify best selling days
Is_Weekend	Weekend indicator	Shopping pattern
Profit_per_Item	Profit / Quantity	Unit profitability
Sales_Category	Sales grouping	Segment orders

The screenshot shows a Jupyter Notebook titled 'Task4.ipynb' in a VS Code environment. The notebook contains three code cells, each defining a new feature from an existing dataset 'df'.

- 1 Profit Ratio**
Code: `df['Profit_Ratio'] = df['Profit'] / df['Sales']`
Purpose: measure profitability of each order.
- 2 Sales Per Item**
Code: `df['Sales_per_Item'] = df['Sales'] / df['Quantity']`
Purpose: approximate product price.
- 3 Discount Amount**
Code: `df['Discount_Amount'] = df['Sales'] * df['Discount']`

The Explorer pane on the left shows the project structure with files like 'Sample - Superstore.csv', 'superstore_feature_engineered.csv', and 'Task4.ipynb'. The bottom status bar indicates 'Spaces: 4', 'Cell 12 of 60', and the date '2/24/2026'.





5. Data Visualization

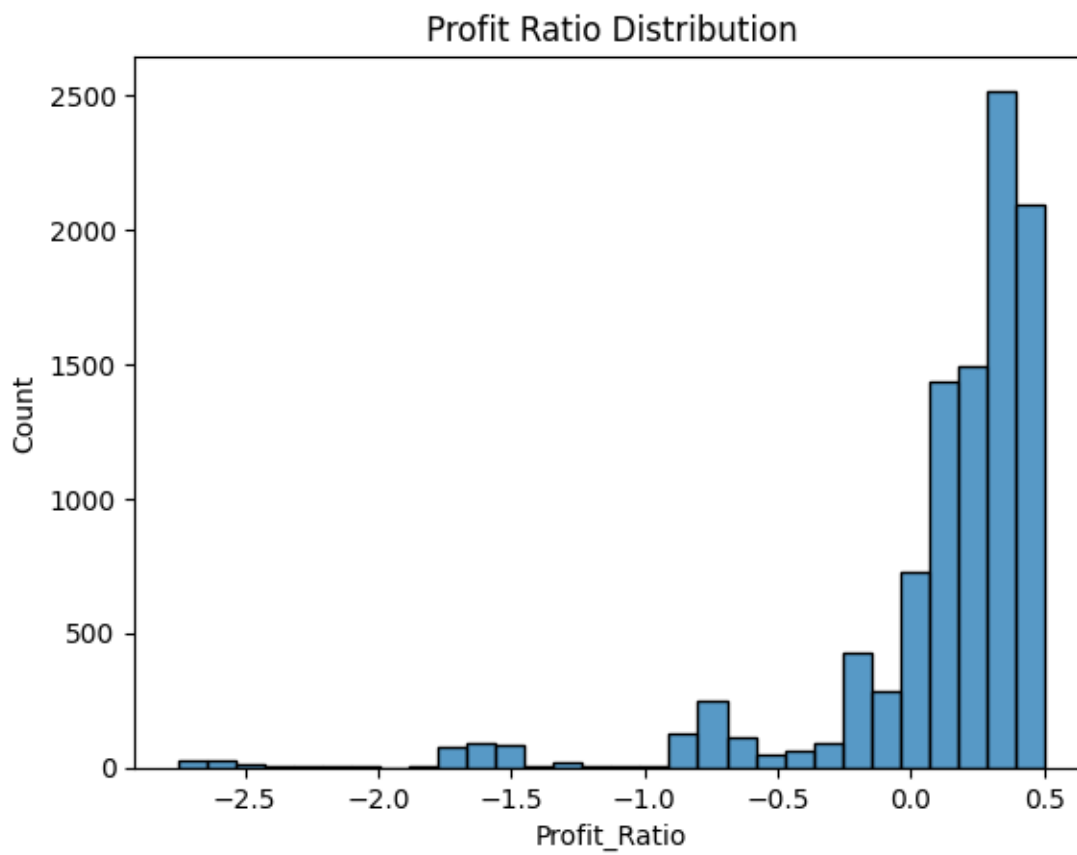
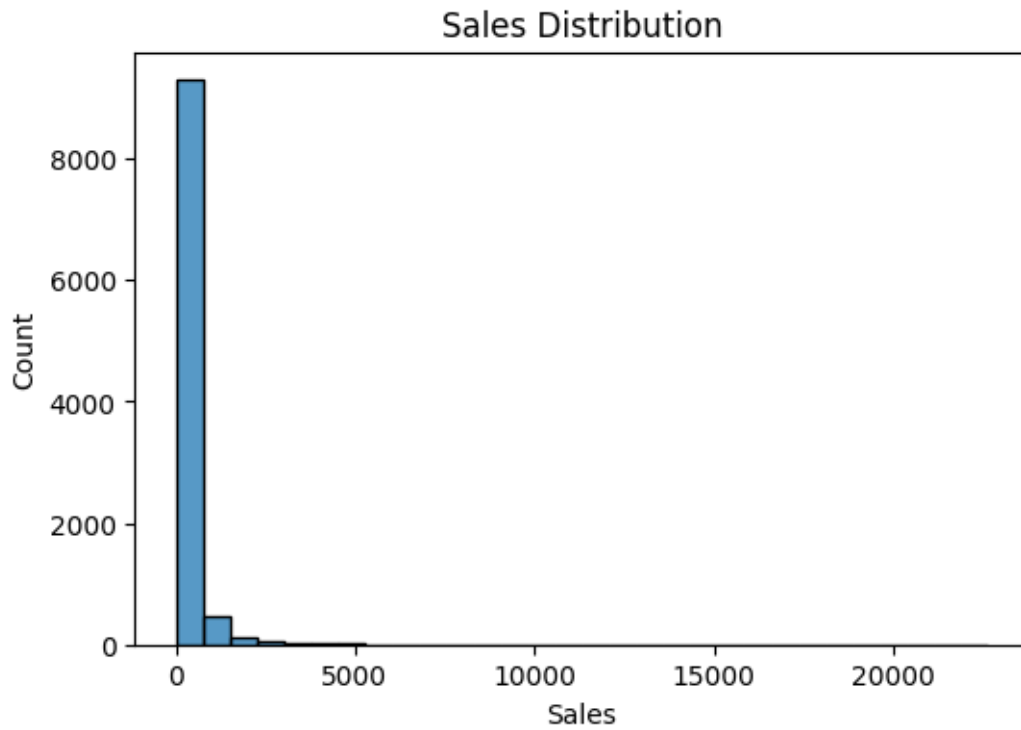
To understand the distribution of the data, several visualizations were created.

Example charts include:

- Sales distribution
- Profit ratio distribution
- Orders by month

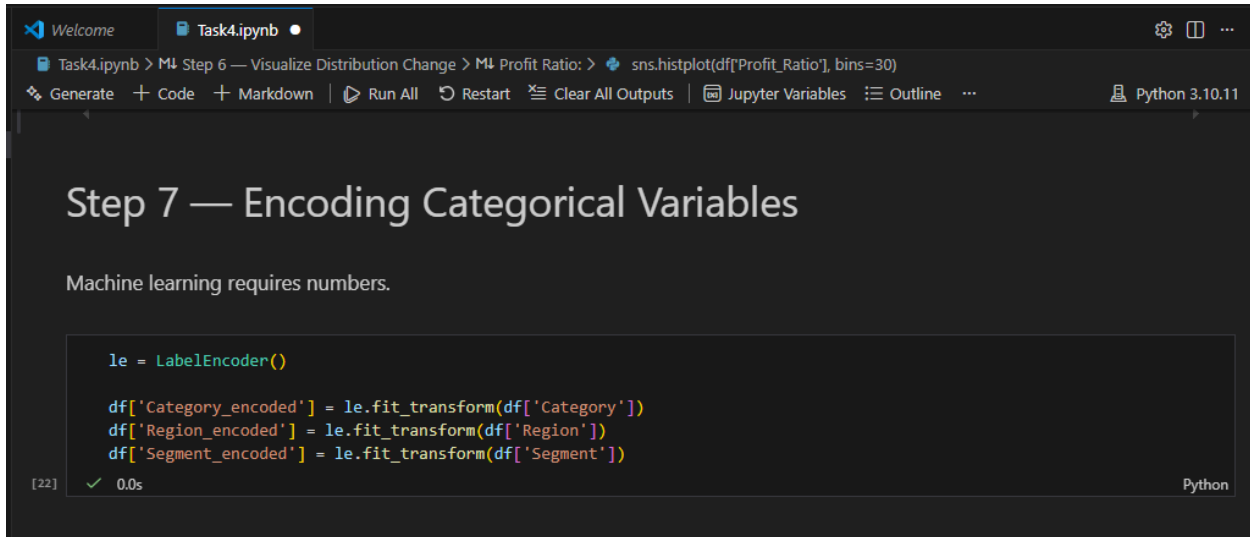
```
sns.histplot(df['Sales'])
```

```
sns.histplot(df['Profit_Ratio'])
```



6. Encoding

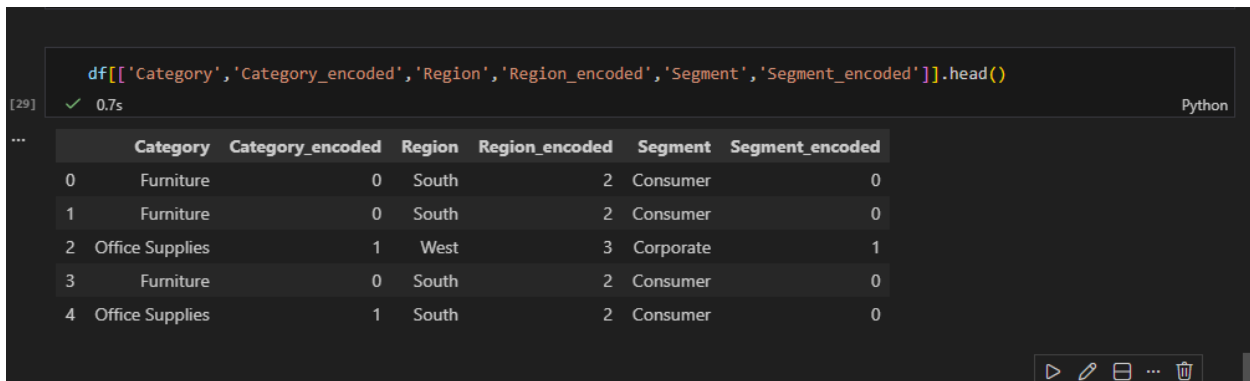
Machine learning algorithms require numerical data. Therefore, categorical variables were encoded using Label Encoding.



```
le = LabelEncoder()

df['Category_encoded'] = le.fit_transform(df['Category'])
df['Region_encoded'] = le.fit_transform(df['Region'])
df['Segment_encoded'] = le.fit_transform(df['Segment'])
```

[22] ✓ 0.0s Python



```
df[['Category', 'Category_encoded', 'Region', 'Region_encoded', 'Segment', 'Segment_encoded']].head()
```

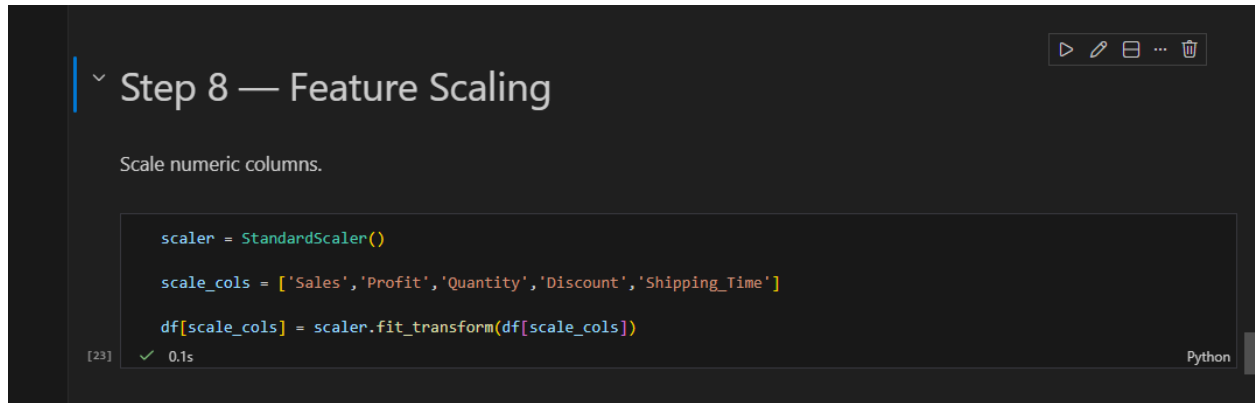
[29] ✓ 0.7s Python

	Category	Category_encoded	Region	Region_encoded	Segment	Segment_encoded
0	Furniture	0	South	2	Consumer	0
1	Furniture	0	South	2	Consumer	0
2	Office Supplies	1	West	3	Corporate	1
3	Furniture	0	South	2	Consumer	0
4	Office Supplies	1	South	2	Consumer	0

7. Feature Scaling

Feature scaling was applied to normalize numerical columns and improve model performance.

StandardScaler was used to scale the data.



The screenshot shows a Jupyter Notebook interface with a dark theme. The cell title is "Step 8 — Feature Scaling". Below the title, the instruction "Scale numeric columns." is displayed. The code cell contains the following Python code:

```
scaler = StandardScaler()

scale_cols = ['Sales', 'Profit', 'Quantity', 'Discount', 'Shipping_Time']

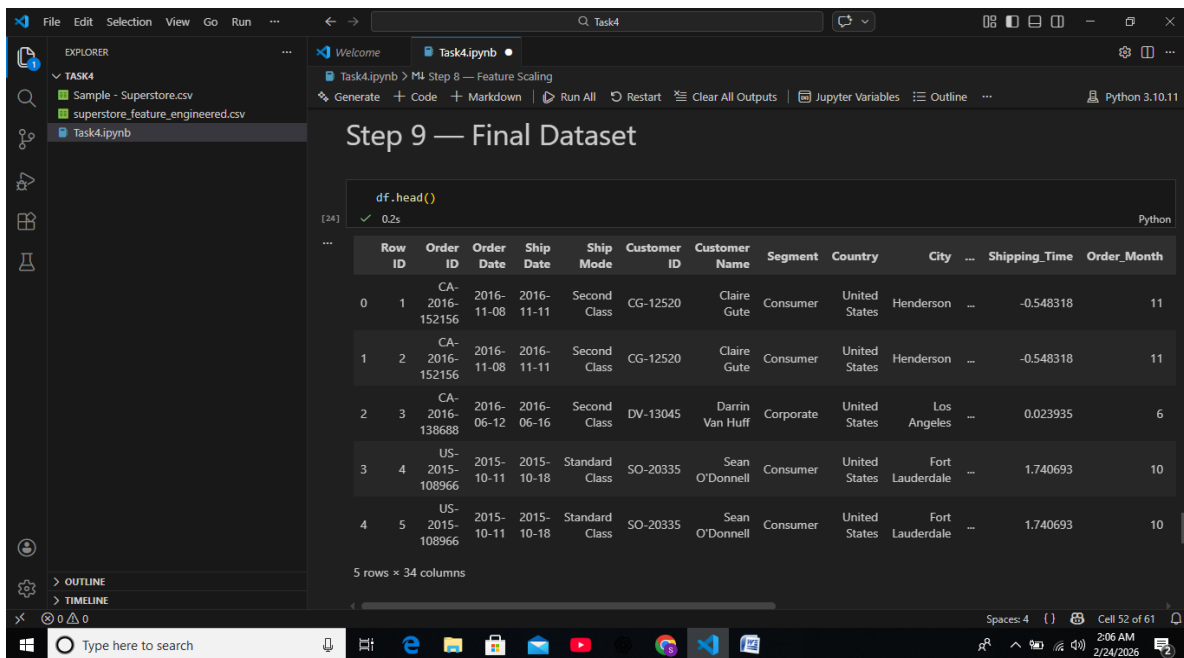
df[scale_cols] = scaler.fit_transform(df[scale_cols])
```

At the bottom left of the code cell, the output is shown as "[23] ✓ 0.1s". At the bottom right, the language is identified as "Python".

8. Final Dataset

After completing feature engineering, encoding, and scaling, the dataset became ready for machine learning and further analysis.

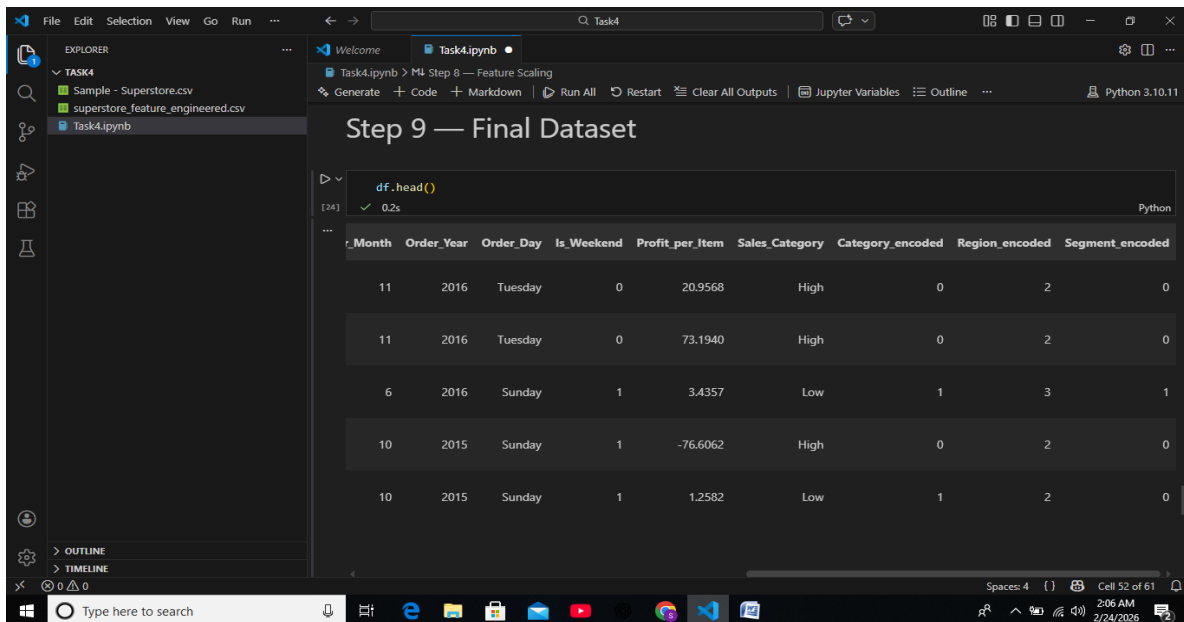
The new dataset contains additional meaningful features that improve analytical capability.



df.head()

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	Shipping Time	Order Month
0	1	2016-11-08	2016-11-11	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	-0.548318	11
1	2	2016-11-08	2016-11-11	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	-0.548318	11
2	3	2016-06-12	2016-06-16	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	0.023935	6
3	4	2015-10-11	2015-10-18	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	1.740693	10
4	5	2015-10-11	2015-10-18	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	1.740693	10

5 rows x 34 columns



df.head()

Order Month	Order Year	Order Day	Is Weekend	Profit per Item	Sales Category	Category_encoded	Region_encoded	Segment_encoded
11	2016	Tuesday	0	20.9568	High	0	2	0
11	2016	Tuesday	0	73.1940	High	0	2	0
6	2016	Sunday	1	3.4357	Low	1	3	1
10	2015	Sunday	1	-76.6062	High	0	2	0
10	2015	Sunday	1	1.2582	Low	1	2	0

Sample - Superstore - Microsoft Excel																			
Row ID																			
Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer	Customer Segment	Country	City	State	Postal Code	Region	Product ID	Category	Sub-Category	Product Name	Sales	Quantity	Discount	Profit
1	CA-2016-1			Second CI	CG-12520	Claire Gut Consumer	United States	Henderson	Kentucky	42420	South	FUR-BO-1	Furniture	Bookcase: Bush Somerset		261.96	2	0	41.9
2	CA-2016-1			Second CI	CG-12520	Claire Gut Consumer	United States	Henderson	Kentucky	42420	South	FUR-CH-1	Furniture	Chairs	Hon Delux	731.94	3	0	219.
3	CA-2016-1			Second CI	DV-13045	Darrin Var Corporate	United States	Los Angeles	California	90036	West	OFF-LA-1	Office Supplies	Labels	Self-Adhesive	14.62	2	0	6.8
4	US-2015-1			Standard	SO-20335	Sean O'Donoghue Consumer	United States	Fort Lauderdale	Florida	33311	South	FUR-TA-1	Furniture	Tables	Bretford C	957.5775	5	0.45	-383.
5	US-2015-1			Standard	SO-20335	Sean O'Donoghue Consumer	United States	Fort Lauderdale	Florida	33311	South	OFF-ST-1	Office Supplies	Storage	Eldon Folio	22.368	2	0.2	2.5
6	CA-2014-1	6/9/2014		Standard	BH-11710	Brosina H Consumer	United States	Los Angeles	California	90032	West	FUR-FU-1	Furniture	Furnishings	Eldon Exp	48.86	7	0	14.1
7	CA-2014-1	6/9/2014		Standard	BH-11710	Brosina H Consumer	United States	Los Angeles	California	90032	West	OFF-AR-1	Office Supplies	Art	Newell 32	7.28	4	0	1.9
8	CA-2014-1	6/9/2014		Standard	BH-11710	Brosina H Consumer	United States	Los Angeles	California	90032	West	TEC-PH-1	Technology	Phones	Mitel 532C	907.152	6	0.2	90.7
9	CA-2014-1	6/9/2014		Standard	BH-11710	Brosina H Consumer	United States	Los Angeles	California	90032	West	OFF-BI-1	Office Supplies	Binders	DXL Angle	18.504	3	0.2	5.7
10	CA-2014-1	6/9/2014		Standard	BH-11710	Brosina H Consumer	United States	Los Angeles	California	90032	West	OFF-AP-1	Office Supplies	Appliance	Belkin F5C	114.9	5	0	34
11	CA-2014-1	6/9/2014		Standard	BH-11710	Brosina H Consumer	United States	Los Angeles	California	90032	West	FUR-TA-1	Furniture	Tables	Chromcraft	1706.184	9	0.2	85.3
12	CA-2014-1	6/9/2014		Standard	BH-11710	Brosina H Consumer	United States	Los Angeles	California	90032	West	TEC-PH-1	Technology	Phones	Konftel 25	911.424	4	0.2	68.3
13	CA-2017-1			Standard	AA-10480	Andrew A Consumer	United States	Concord	North Carolina	28027	South	OFF-PA-1	Office Supplies	Paper	Xerox 196	15.552	3	0.2	5.4
14	CA-2016-1			Standard	IM-15070	Irene Mac Consumer	United States	Seattle	Washington	98103	West	OFF-BI-1	Office Supplies	Binders	Fellowes	407.976	3	0.2	132.5
15	US-2015-1			Standard	HP-14815	Harold Pa Home Office	United States	Fort Worth	Texas	76106	Central	OFF-AP-1	Office Supplies	Appliance	Holmes R	68.81	5	0.8	-123.
16	US-2015-1			Standard	HP-14815	Harold Pa Home Office	United States	Fort Worth	Texas	76106	Central	OFF-BI-1	Office Supplies	Binders	Storex Duo	2.544	3	0.8	-3.
17	CA-2014-1			Standard	PK-19075	Pete Kriz Consumer	United States	Madison	Wisconsin	53711	Central	OFF-ST-1	Office Supplies	Storage	Stur-D-Stc	665.88	6	0	13.3
18	CA-2014-1			Second CI	AG-10270	Alejandro Consumer	United States	West Jordan	Utah	84084	West	OFF-ST-1	Office Supplies	Storage	Fellowes	55.5	2	0	5
19	CA-2014-1	9/1/2014		Second CI	ZD-21925	Zuschuss I Consumer	United States	San Francisco	California	94109	West	OFF-AR-1	Office Supplies	Art	Newell 34	8.56	2	0	2.4
20	CA-2014-1	9/1/2014		Second CI	ZD-21925	Zuschuss I Consumer	United States	San Francisco	California	94109	West	TEC-PH-1	Technology	Phones	Cisco SPA	213.48	3	0.2	16.
21	CA-2014-1	9/1/2014		Second CI	ZD-21925	Zuschuss I Consumer	United States	San Francisco	California	94109	West	OFF-BI-1	Office Supplies	Binders	Wilson Jones	22.72	4	0.2	7.
22	CA-2016-1			Standard	KB-16585	Ken Black Corporate	United States	Fremont	Nebraska	68025	Central	OFF-AR-1	Office Supplies	Art	Newell 31	19.46	7	0	5.0
23	CA-2016-1			Standard	KB-16585	Ken Black Corporate	United States	Fremont	Nebraska	68025	Central	OFF-AP-1	Office Supplies	Appliance	Acco Six-C	60.34	7	0	15.6
24	US-2017-1			Second CI	SF-20065	Sandra Fla Consumer	United States	Philadelphia	Pennsylvania	19140	East	FUR-CH-1	Furniture	Chairs	Global Deluxe	71.372	2	0.3	-1.0

superstore_feature_engineered - Microsoft Excel																								
Row ID																								
	O	P	Q	R	S	T	U	V	W	X	Y	Z	AA	AB	AC	AD	AE	AF	AG	AH	AI			
1	Category	Sub-Category	Product Name	Sales	Quantity	Discount	Profit	Profit_Ratio	Sales_per_	Discount	Shipping_Order	Mc Order_Ye	Order_Dat	Is_Week	Profit_per	Sales_Cat	Category	Region	er	Segment	encod			
2	Furniture	Bookcase: Bush	Som	0.05151	-0.8043	-0.75664	0.056593	0.16	130.98	0	-0.54832	11	2016	Tuesday	0	20.9568	High	0	2	0				
3	Furniture	Chairs	Hon Delu	0.805633	-0.35486	-0.75664	0.815054	0.3	243.98	0	-0.54832	11	2016	Tuesday	0	73.194	High	0	2	0				
4	Office Su	Labels	Self-Adhe	-0.34537	-0.8043	-0.75664	-0.093	0.47	7.31	0	0.023935	6	2016	Sunday	1	3.4357	Low	1	3	1				
5	Furniture	Tables	Bretford C	1.167688	0.544012	1.423149	-1.75748	-0.4	191.5155	430.9099	1.740693	10	2015	Sunday	1	-76.6062	High	0	2	0				
6	Office Su	Storage	Eldon Folk	-0.33294	-0.8043	0.212153	-0.11159	0.1125	11.184	4.4736	1.740693	10	2015	Sunday	1	1.2582	Low	1	2	0				
7	Furniture	Furnishing	Eldon Exp	-0.29043	1.442889	-0.75664	-0.06185	0.29	6.98	0	0.596187	6	2014	Monday	0	2.0242	Medium	0	3	0				
8	Office Su	Art	Newell 32	-0.35715	0.094574	-0.75664	-0.11394	0.27	1.82	0	0.596187	6	2014	Monday	0	0.4914	Low	1	3	0				
9	Technolog	Phones	Mitel 532	1.086776	0.993451	0.212153	0.264925	0.1	151.192	181.4304	0.596187	6	2014	Monday	0	15.1192	High	2	3	0				
10	Office Su	Binders	DXL Angle	-0.33914	-0.35486	0.212153	-0.09765	0.3125	6.168	3.7008	0.596187	6	2014	Monday	0	1.9275	Low	1	3	0				
11	Office Su	Appliance	Belkin F5C	-0.18446	0.544012	-0.75664	0.024816	0.3	22.98	0	0.596187	6	2014	Monday	0	6.894	Medium	1	3	0				
12	Furniture	Tables	Chromcam	2.368891	2.341766	0.212153	0.241847	0.05	189.576	341.2368	0.596187	6	2014	Monday	0	9.4788	High	0	3	0				
13	Technolog	Phones	Konftel 25	1.093631	0.094574	0.212153	0.169478	0.075	227.856	182.2848	0.596187	6	2014	Monday	0	17.0892	High	2	3	0				
14	Office Su	Paper	Xerox 196	-0.34387	-0.35486	0.212153	-0.0991	0.35	5.184	3.1104	0.596187	4	2017	Saturday	1	1.8144	Low	1	2	0				
15	Office Su	Binders	Fellowes 1	0.285806	-0.35486	0.212153	0.443697	0.325	135.992	81.5952	0.596187	12	2016	Monday	0	44.1974	High	1	3	0				
16	Office Su	Appliance	Holmes Ru	-0.25842	0.544012	3.118544	-0.65108	-1.8	13.762	55.048	0.023935	11	2015	Sunday	1	-24.7716	Medium	1	0	2				
17	Office Su	Binders	Storex Du	-0.36474	-0.35486	3.118544	-0.13863	-1.5	0.848	2.0352	0.023935	11	2015	Sunday	1	-1.272	Low	1	0	2				
18	Office Su	Storage	Stur-D-Stc	0.699635	0.993451	-0.75664	-0.06548	0.02	110.98	0	1.740693	11	2014	Tuesday	0	2.2196	High	1	0	0				
19	Office Su	Storage	Fellowes 1	-0.27977	-0.8043	-0.75664	-0.07969	0.18	27.75	0	-1.12057	5	2014	Tuesday	0	4.995	Medium	1	3	0				
20	Office Su	Art	Newell 34	-0.35509	-0.8043	-0.75664	-0.11174	0.29	4.28	0	0.596187	8	2014	Wednesd	0	1.2412	Low	1	3	0				
21	Technolog	Phones	Cisco SPA	-0.02628	-0.35486	0.212153	-0.05398	0.075	71.16	42.696	0.596187	8	2014	Wednesd	0	5.337	High	2	3	0				
22	Office Su	Binders	Wilson Joi	-0.33237	0.094574	0.212153	-0.09081	0.325	5.68	4.544	0.596187	8	2014	Wednesd	0	1.846	Low	1	3	0				
23	Office Su	Art	Newell 31	-0.3376	1.442889	-0.75664	-0.10074	0.26	2.78	0	0.023935	12	2016	Friday	0	0.7228	Low	1	0	1				
24	Office Su	Appliance	Acco Six-C	-0.27201	1.442889	-0.75664	-0.05536	0.26	8.62	0	0.023935	12	2016	Friday	0	2.2412	Medium	1	0	1				
25	Furniture	Chairs	Global De	-0.2543	-0.8043	0.696552	-0.12669	-0.01429	35.686	21.4116	-1.12057	7	2017	Sunday	1	-0.5098	Medium	0	1	0				

9. Conclusion

In this task, feature engineering and transformation techniques were successfully applied to the Superstore dataset.

Key achievements:

- Created multiple new features
- Performed encoding of categorical variables
- Applied feature scaling
- Prepared dataset for machine learning

These steps improve the quality of the dataset and help build more accurate predictive models.